THE CONSTRUCTIBLE SETS ARE CLOSED UNDER FINITE INTERSECTIONS, FINITE UNIONS AND COMPLEMENTS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The definition of a constructible set as a finite union of locally closed sets, and the notion of a retrocompact open (an open whose inclusion is quasi-compact). Set up in the author's section on constructible sets, immediately before the counted statement. Fixes the family whose closure properties the statement asserts and the notion of retrocompactness used in the proof.

Lemma 0.1. *The collection of constructible sets is closed under finite intersections, finite unions and complements.*

Proof. Note that if U_1, U_2 are open and retrocompact in X then so is $U_1 \cup U_2$ because the union of two quasi-compact subsets of X is quasi-compact. It is also true that $U_1 \cap U_2$ is retrocompact. Namely, suppose $U \subset X$ is quasi-compact open, then $U_2 \cap U$ is quasi-compact because U_2 is retrocompact in X, and then we conclude $U_1 \cap (U_2 \cap U)$ is quasi-compact because U_1 is retrocompact in X. From this it is formal to show that the complement of a constructible set is constructible, that finite unions of constructibles are constructible, and that finite intersections of constructibles are constructible. □

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